

A New Approach to Radiation Therapy Treatment Planning Problems Using Machine Learning: Model Implementation

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Abstract—Radiotherapy is a critical treatment for cancer patients, requiring precise delivery of radiation doses to maximize effectiveness and minimize side effects. However, conventional methods for determining dosages often lead to suboptimal outcomes due to expert-driven guidelines and laborious linear programming approaches. This research seeks to explore alternative calculation methods to provide optimal dosages for radiotherapy, focusing on external beam radiotherapy. In this technique, radiation is emitted from a rotating device targeting the tumor or affected area. To protect healthy tissue, the region is divided into 4 by 4 by 4-millimeter voxels, enabling detailed radiation dosage simulations. Machine learning (ML) techniques have shown promise in medical applications, and this research aims to leverage ML algorithms to optimize radiotherapy dosage calculations. A neural network model, comprising interconnected neurons, is trained on patient data to learn optimal dosage patterns. Evaluation is done using Root Mean Squared Error (RMSE) and Mean Squared Error (MSE) metrics. Preliminary results demonstrate significant improvements in dosage optimization, achieving a remarkable mean MSE reduction from 5.2938 to 0.0004 after 160 iterations. The goal of future results is to populate a Dose Volume Histogram (DVH) using Computational Environment for Radiotherapy Treatment (CERR) to showcase how well the ML model would perform in an actual radiotherapy planning setting. This research shows the potential to revolutionize radiotherapy, offering personalized and efficient dosages while reducing complications. By harnessing machine learning algorithms, this research aims to enhance treatment efficacy and improve patient outcomes in cancer radiotherapy. Ultimately, this approach may lead to more effective and targeted treatments, transforming the landscape of cancer care.

I. INTRODUCTION

Radiotherapy is a critical and widely used treatment option for cancer patients, where precise delivery of radiation doses is crucial for maximizing therapeutic outcomes while minimizing side effects. Cancer has a significant impact on countless lives, with over 1.6 million cases and 600,000 deaths reported in the year 2020 alone [1]. Improving the treatment of this disease could have a profound positive effect on a vast number of people. Traditionally, determining the dosage for radiotherapy relied on expert-driven guidelines and heuristics, often resulting in suboptimal dosages for individual patients. The current method, which involves linear programming and constraints, requires adjusting numerous variables for each

patient, leading to high costs, inefficiency, and occasional inaccuracies. This research aims to explore alternative calculation methods to supply optimal dosages for radiotherapy treatment, moving away from linear programming.

The specific form of radiotherapy under focus in this research is external beam radiotherapy. This technique involves emitting radiation from a device with a moving head that rotates around the patient, targeting the tumor or the affected region [2]. The purpose is to maximize the exposure time to the tumor while minimizing direct exposure to the surrounding healthy tissue that the radiation must pass through to reach the tumor. Radiation effectively damages cell structures, making it useful for targeting tumor cells, but it can also harm healthy cells that the body requires for normal functioning. Without precise targeting, the healthy tissue would receive the same lethal doses of radiation as the tumor, significantly impacting the patient's quality of life and potentially leading to organ failures. To address this, the research divides the affected area into 4 by 4 by 4-millimeter cubes called voxels, allowing for radiation dosage calculations throughout the region based on complex computer simulations and models due to the lack of feasible data gathering options.

In recent years, machine learning techniques have demonstrated promise in various medical applications. From image generation to plant maturity detection, machine learning holds potential for determining radiation levels in radiotherapy [3]. By leveraging the wealth of cancer-related data available, a machine learning model can be trained to achieve higher accuracy. While current optimizations rely on linear programming, machine learning has the potential to optimize radiotherapy treatment more efficiently and with greater success. This research proposal aims to explore the utilization of machine learning algorithms to optimize dosage calculations in radiotherapy treatment, with the ultimate goal of improving treatment efficacy and patient outcomes.

Radiotherapy plays a pivotal role in cancer treatment, delivering targeted radiation to destroy or inhibit the growth of tumor cells. However, customizing optimal dosages for individual patients can be complex, considering the interplay of tumor characteristics, patient-specific factors, and radiation-induced side effects. Machine learning (ML) algorithms offer a promising solution to address these challenges by analyzing extensive datasets, identifying patterns, and optimizing dosages based on individual patient profiles. The proposed research aims to develop and evaluate a machine learning framework for optimizing dosage calculations in radiotherapy treatment, ultimately providing patients with the most effective treatment while minimizing any negative impact on their quality of life, without the need for manual adjustments of variables.

II. METHODS

1) *Preparation to Create the ML Model:* Utilizing a desktop computer equipped with an Nvidia Quadro P5000 GPU as the main processor, a machine learning model was developed. The software of choice for this endeavor was MATLAB, a widely used programming environment known for its capabilities in scientific computing and machine learning applications. The input data required for the model was provided in the form of a 2-dimensional [1 500] matrix containing the intensities of the beamlets of radiation. Each row of the matrix represented a distinct instance with relevant features or attributes stored in the columns. The output data required for the model was provided in the form of a 2-dimensional [1 500] matrix containing levels of radiation in each voxel, reflecting spatial distribution in a 3D context. Voxels are used in medical imaging to depict parts of a patient's body in a three-dimensional picture. Medical experts may examine and study interior structures in three dimensions because each voxel in a medical picture provides information about the tissue or substance present at that particular spot. For experimental purposes, the data was sourced from the MATRAD open-source database, which was then processed through a Monte-Carlo simulation to obtain the appropriate output data for the ML model.

2) *Creation of the ML Model:* With the structural code of the model provided for a basic neural network model that generates results using randomly generated values, the code was modified to accept true patient data. The program was modified to split the given data in an 80:20 ratio. 80% of the dataset was used to train the model while the other 20% of the dataset was used to test the model. The model consists of several layers with thousands of neurons on each layer. A neural network model has many layers, each of which extracts higher-level characteristics from previously acquired representations. The deeper it goes into the network; this hierarchical approach enables the model to learn increasingly abstract and sophisticated properties. The first layers may catch basic patterns like edges and textures, while the deeper layers pick up more complex patterns, enabling the model to comprehend complex data structures. Neural network models have the capacity to derive data representations from the original input. As opposed to standard machine learning techniques, this eliminates the requirement for manual feature engineering. Neural network models may more effectively generalize to novel and unexplored data by automatically learning meaningful representations at various levels.

3) *Calculating RMSE:* Root Mean Squared Error is referred to as RMSE. It is a widely used statistic to assess a predictive model's precision, especially in regression applications [4]. RMSE measures the discrepancy between a dataset's actual (observed) values and anticipated values. By using the square root of the mean of the squared errors, it calculates the average magnitude of the errors between the predicted values and the true values. The formula for calculating RMSE is provided in Figure 1. The skeleton neural network model, however, already implement calculation of RMSE in the program.

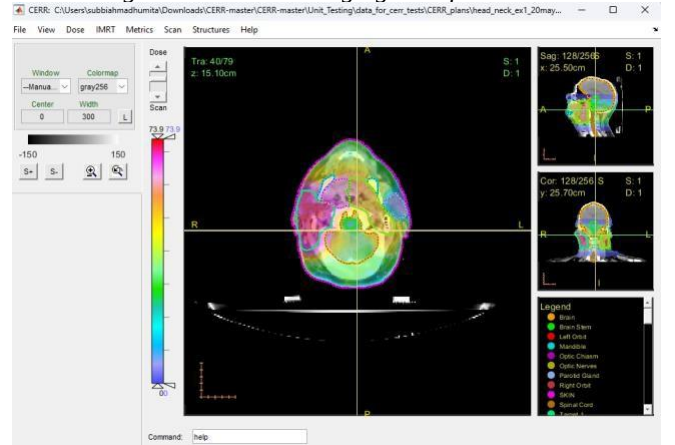
Figure 1: Formula for Root Mean Squared Error [5]

$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$$

4) *Calculating Loss:* Neural network models have the capacity to learn data representations directly from the input's raw data. With this, there is less need for human feature engineering, which is frequently necessary with standard machine learning techniques. Neural network models are better able to generalize to new and unexplored data since they automatically develop meaningful representations at many levels [6]. Mean squared error (MSE) loss function that deals with regression task with continuous data. Between the projected values and the actual target values, it calculates the average squared difference. To enhance its performance on the given task, the model iteratively adjusts its parameters during training to minimize the selected loss function. Calculating the gradients of the loss with respect to the model's parameters is a common step in the optimization process. Using these gradients, the parameters are subsequently updated in a way that minimizes the loss. Until the model reaches an acceptable level of performance or converges to a stable state, this iterative process is continued.

5) *Using CERR to Populate DVH Curves:* CERR is a Computational Environment for Radiotherapy Research. An open-source software platform called CERR was created to support radiation researchers and practitioners. For processing, analyzing, and displaying medical imaging data linked to radiation treatment planning and research, it offers several tools and features as shown in Figure 2. The Digital Imaging and Communications in Medicine (DICOM) files, the RT-STRUCT (structure set) files, the RT-DOSE (dose distribution) files, and the RT-PLAN (treatment plan) files are only a few of the several data types that CERR supports [7].

Figure 2: Medical imaging data from CERR



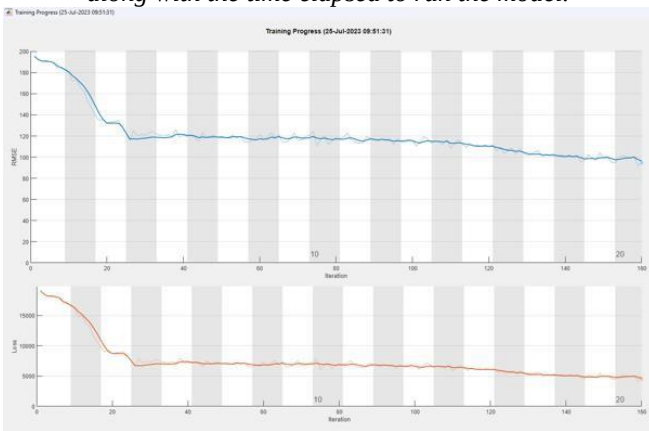
6) Users may import and handle different data types from radiation treatment planning systems with the use of these features. For contouring target structures (organs, tumors, etc.), computing dose-volume histograms (DVH) (discussed in later section), and conducting dose

calculations, the program provides a complete range of capabilities. To provide the best possible treatment planning, it enables researchers and physicians to display and evaluate dose distributions, isodose curves, and other pertinent data. Additionally, CERR interfaces with the MATLAB environment, giving users a wealth of customization possibilities and the freedom to create and apply bespoke algorithms and research methodologies.

III. RESULTS/DISCUSSION

1) *Current Results:* The current ML model has achieved an MSE of 0.0004. With an initial calculation of 5.2938. With each trial, the resulting MSE has ranged between 0.0004 to 0.0010 with average running the program in 8 seconds. Figure 3 shows the RMSE vs Iterations graphs and the Loss vs Iterations. For testing purposes, 160 iterations of the model were run.

Figure 3: RMSE vs Iterations and Loss vs Iterations graphs along with the time elapsed to run the model.



2) *DVH:* The tool commonly utilized by physicians to evaluate treatment plan effectiveness is known as the (cumulative) dose-volume histogram (DVH). This histogram provides information about the fraction of a given target or critical structure's volume that receives at least a specific dose amount [8]. By representing structures using voxels, the DVH value at a certain dose, d , precisely indicates the fraction of voxels within the structure that receive a dose of d units or higher.

For example, in Figure 4, the DVH point for the target (indicated by the right vertical dotted line) indicates that approximately 95% of the target volume receives at least 73.8 units of dose, measured in Gray (Gy). Similarly, the DVH point for the critical structure (indicated by the left vertical dotted line) reveals that about 26% of the critical structure receives at least 30 Gy.

The DVH curve from this data set was populated using CERR to showcase well the neural network model would perform in an actual radiotherapy planning setting. The DVH curves were plotted for all structures, as shown in Figure 5, with the most updated version of the neural network model.

Figure 4: Example of DVH for a target and critical structure [9]

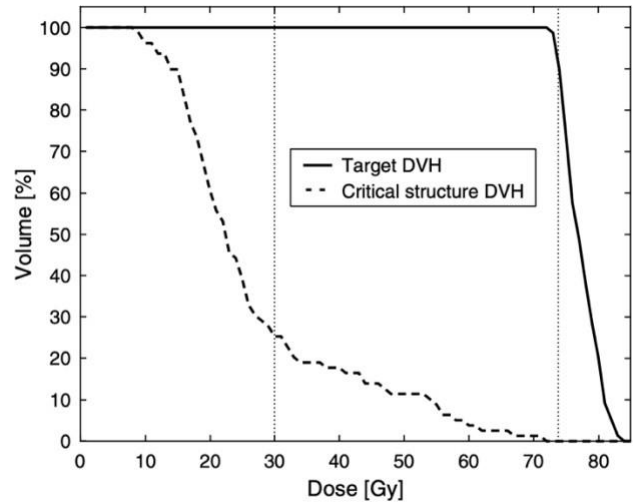
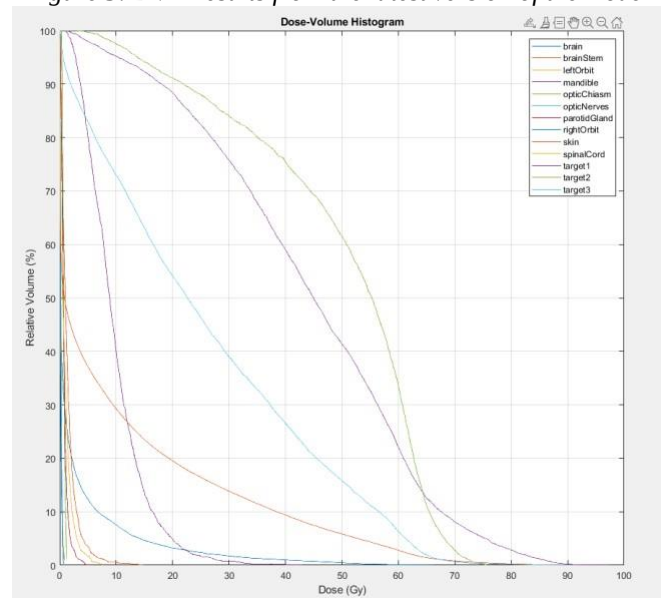


Figure 5: DVH results from the latest version of the model



3) *Expected Results:* The main goal of a neural network model is to reduce the RMSE or the selected loss function. The predictions of the model are said to be closer to the genuine target values in the training data when the RMSE or loss values are lower. With increased training of the model, a lower loss and RMSE should be achievable. The accuracy of the model's predictions is increased when RMSE or loss are minimized. A low RMSE or loss indicates that the model is more accurate and dependable since it is correctly predicting the target values with fewer errors. The model is effectively fitting the training set of data when the RMSE or loss is minimal. For the model to generalize to previously unknown data, it is essential that the model can recognize and understand the underlying patterns and connections in the data. Finding the model parameters (weights and biases) that minimize the selected loss function is the aim when training a neural network model. To accomplish this goal, training entails employing optimization techniques to iteratively update these parameters.

4) *Limitations*: When optimizing the model, it's crucial to find a balance, though. Overfitting is a problem when the model gets too complicated and tends to memorize the training data, which causes it to perform poorly on unobserved data. Overfitting may be reduced, and the model's generalizability can be ensured via regularization approaches and performance monitoring on validation data.

IV. CONCLUSION

In conclusion, this research endeavors to revolutionize radiotherapy treatment by exploring alternative calculation methods for optimal dosage delivery. Radiotherapy is a critical and widely used treatment option for cancer patients, and precise radiation dose administration is essential to maximize therapeutic benefits while minimizing harmful side effects. Traditional approaches relying on expert-driven guidelines and linear programming often result in suboptimal dosages, leading to potential inefficiencies and inaccuracies.

By delving into machine learning algorithms, this study seeks to harness the potential of extensive cancer-related data to develop an ML framework that optimizes radiation dosage calculations. The proposed neural network model, trained on patient data and evaluated through Root Mean Squared Error (RMSE) and Mean Squared Error (MSE) metrics, demonstrates promising results in reducing dosage errors and improving accuracy.

The focus on external beam radiotherapy and the subdivision of the affected area into voxels allows for complex computer simulations, providing precise calculations for radiation dosages. This approach is a significant departure from the conventional linear programming methods and offers the potential to streamline treatment planning, reducing costs and minimizing unnecessary adjustments of variables for individual patients. The preliminary outcomes signify the potential transformative impact of ML-driven optimization in radiotherapy treatment. Achieving a substantial reduction in MSE from 5.2938 to 0.0004 after 160 iterations highlights the effectiveness of the proposed approach.

The implications of this research extend beyond improved treatment efficacy. By leveraging machine learning to customize dosages, patients can benefit from personalized treatment plans that enhance the overall quality of life during and after radiotherapy. Moreover, the reduction of treatment-related complications holds the promise of minimizing long-term adverse effects and contributing to better patient outcomes.

The utilization of machine learning algorithms for optimizing radiation dosage calculations in radiotherapy presents a valuable avenue for advancing cancer treatment. By empowering clinicians with data-driven insights, this research aims to enhance treatment precision, efficacy, and

patient well-being, potentially transforming the landscape of cancer care. As research progresses, the prospect of implementing this ML-driven approach in clinical practice holds the potential to positively impact the lives of countless cancer patients, offering new hope for a brighter and healthier future.

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